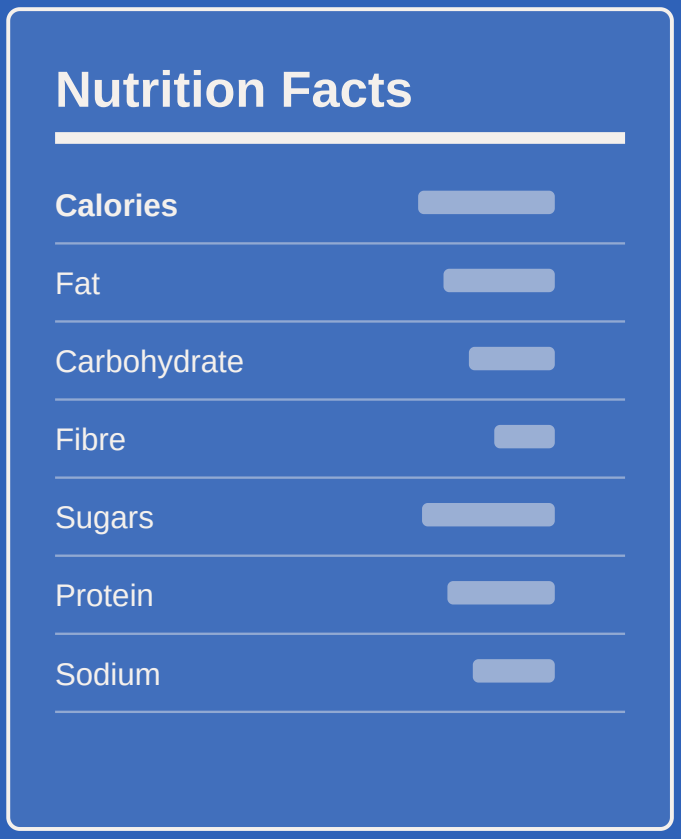


The Recipe Vault · Vol. 03

Know Your Food

Labels, nutrients, portions, safety

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CHAPTER

Energy and calories

Food gives your body energy, measured in **calories** (strictly, kilocalories, written kcal or Cal). Every food label in Canada and the US shows calories per serving.

Where calories come from

NUTRIENT	CALORIES PER GRAM
Carbohydrate	4
Protein	4
Fat	9
Alcohol	7
Fibre	about 2 (partly digested)

That's why a tablespoon of oil (about 14 g of fat) has around 120 calories, while a whole cup of broccoli has about 30.

How much energy do you need?

It depends on your age, size, sex and how active you are. Many adults need somewhere between 1,800 and 3,000 calories a day. The **% Daily Value** on Canadian labels is based on a 2,000-calorie diet, as a reference point, not a target for everyone.

You don't need to count calories to eat well. Building balanced meals (chapter 4) and noticing hunger and fullness works for most people. Counting can help if you have a specific goal; if it makes you anxious about food, it isn't worth it.

Calories aren't the whole story

A 200-calorie snack of nuts and a 200-calorie soft drink have the same energy but very different effects. The nuts bring protein, fibre, healthy fats and minerals that keep you full. The soft drink brings sugar and little else. **What a food contains** matters as much as how much energy it has.

CHAPTER

The big three: macronutrients

Protein

Protein builds and repairs muscles, skin, hair, enzymes and hormones. It also keeps you full longer than carbs or fat.

- **Sources:** meat, poultry, fish, eggs, dairy, tofu, tempeh, paneer, beans, lentils, chickpeas, nuts and seeds.
- **How much:** a common guide for healthy adults is about **0.8 g per kg of body weight per day** (about 56 g for a 70 kg adult). Older adults, active people and those recovering from illness often benefit from more.
- **Easy target:** include a palm-sized portion of protein at each meal.

Carbohydrates

Carbs are your body's preferred quick fuel, especially for the brain and for exercise.

- **Whole carbohydrates** come with fibre, vitamins and minerals: whole grains, oats, brown rice, beans, lentils, fruit, vegetables.
- **Refined carbohydrates** have had much of that removed: white bread, pastries, sweets, sugary drinks.
- **Sugars** occur naturally in fruit and milk, and are added to many packaged foods. Health Canada and many health groups advise limiting added and "free" sugars.

Fibre

Fibre is a carbohydrate you don't digest. It keeps your gut healthy, helps with regularity, slows the rise in blood sugar after meals and helps you feel full.

Health Canada's recommended intake is about **25 g a day for adult women and 38 g for adult men**. Most people eat far less.

Easy ways to get more: oats for breakfast, beans or lentils in soups and salads, whole-grain bread, fruit with the skin on, a handful of nuts, an extra vegetable at dinner.

Fats

Fat carries flavour, helps you absorb vitamins A, D, E and K, and is essential for your brain and cells.

TYPE	FOUND IN	CHOOSE
Unsaturated (mono and poly)	Olive oil, canola oil, avocado, nuts, seeds, fatty fish	**Most of the time**
Omega-3 (a polyunsaturated fat)	Salmon, sardines, trout, walnuts, chia, flax	**Often**
Saturated	Butter, cheese, fatty meat, coconut oil, palm oil	**In smaller amounts**
Trans	Industrially produced partially hydrogenated oils	**Avoid.** Canada banned them in foods in 2018

CHAPTER

Vitamins, minerals and water

Micronutrients are needed in small amounts, but you can't live without them. A varied diet with plenty of colour covers most needs.

NUTRIENT	WHAT IT DOES	GOOD SOURCES
Vitamin A	Vision, immunity, skin	Carrots, sweet potato, spinach, eggs, milk
B vitamins (B6, B12, folate)	Energy, red blood cells, nerves	Whole grains, meat, eggs, legumes, leafy greens. **B12 only in animal foods or fortified foods**
Vitamin C	Immunity, healing, iron absorption	Peppers, citrus, strawberries, broccoli, kiwi
Vitamin D	Bones, immunity	Fatty fish, fortified milk and plant drinks, egg yolks. Hard to get from sun in a Canadian winter
Vitamin K	Blood clotting, bones	Leafy greens, broccoli, Brussels sprouts
Calcium	Bones, teeth, muscles	Milk, yogurt, cheese, fortified plant drinks, tofu made with calcium, canned salmon with bones
Iron	Carries oxygen in the blood	Red meat, chicken, beans, lentils, tofu, fortified cereals. Eat with vitamin C for better absorption
Potassium	Heart, muscles, blood pressure	Potatoes, bananas, beans, yogurt, leafy greens
Magnesium	Muscles, nerves, energy	Nuts, seeds, whole grains, leafy greens, dark chocolate
Zinc	Immunity, healing	Meat, shellfish, legumes, seeds, dairy
Iodine	Thyroid hormones	Iodized salt, dairy, seafood, eggs
Sodium	Fluid balance	Salt, and most packaged and restaurant food. **Most people eat too much**

About supplements: most healthy people can get what they need from food. Some groups may need supplements, for example vitamin D in winter, folic acid before and during pregnancy, and B12 for vegans. Ask a doctor or pharmacist before starting any supplement.

Water

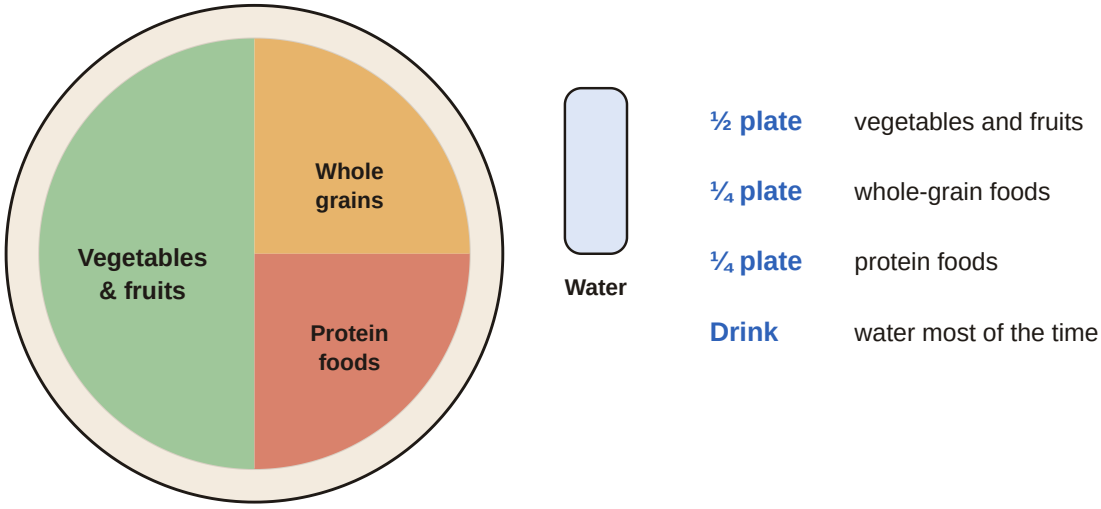
Your body is about 60% water. Needs vary, but a common guide is around **2–3 litres of fluids a day** for adults, including water from food. Drink more in hot weather and when exercising.

- Make **water your drink of choice**. Plain milk and unsweetened plant drinks are good too.
- Pale yellow urine is a simple sign you're drinking enough.
- Juice, soft drinks and many coffee-shop drinks contain a lot of sugar.

CHAPTER

Building a balanced plate

Canada's Food Guide suggests a simple picture for most meals:



½ plate	vegetables and fruits
¼ plate	whole-grain foods
¼ plate	protein foods
Drink	water most of the time

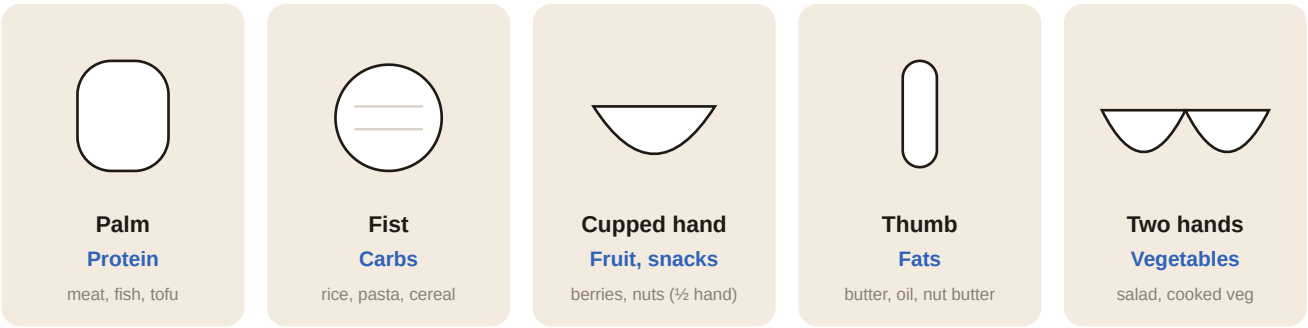
The balanced plate, based on Canada's Food Guide.

- **Half the plate:** vegetables and fruits, in a range of colours.
- **A quarter:** whole-grain foods such as brown rice, whole-wheat pasta, oats, quinoa or whole-grain bread.
- **A quarter:** protein foods, choosing plant-based proteins more often.
- **To drink:** water.

This isn't a strict rule. A curry, a stir-fry or a sandwich can follow the same proportions without being on a plate.

Portion sizes with your hand

Your hand is always with you, and it's sized to your body.



Quick portion guide using your hand. Adjust for your appetite and activity.

Glycemic index, in plain words

The **glycemic index (GI)** describes how quickly a carbohydrate food raises blood sugar.

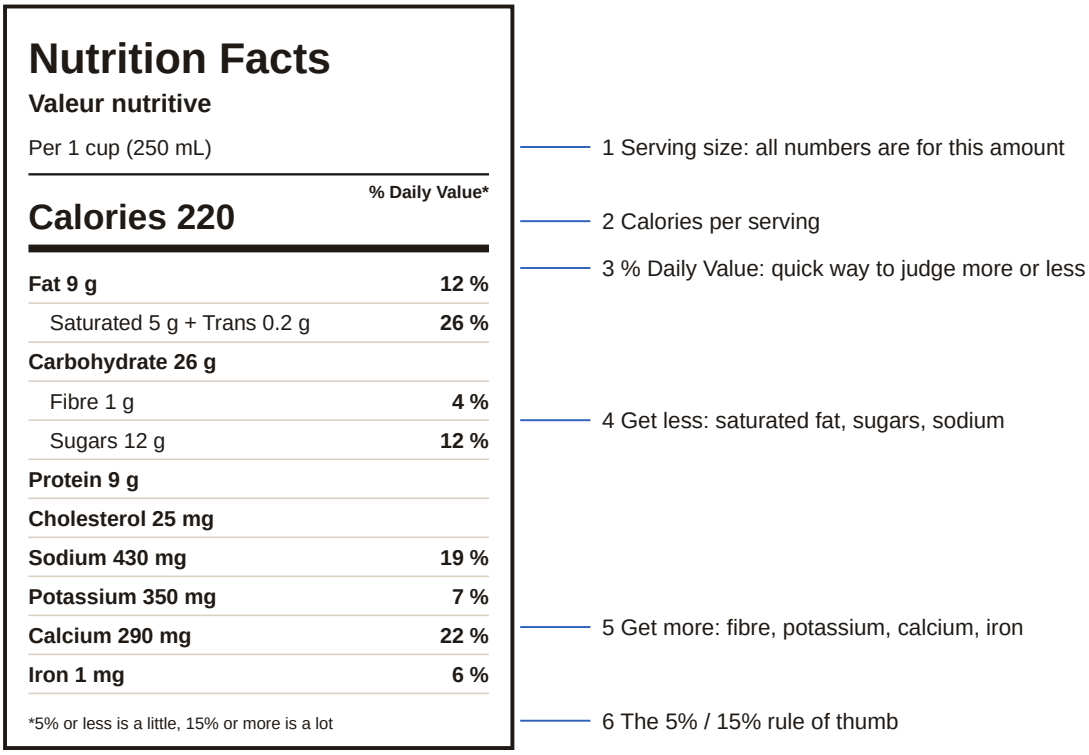
- **Lower GI:** most beans and lentils, oats, barley, most fruit, whole-grain and sourdough breads, pasta cooked al dente.
- **Higher GI:** white bread, most breakfast cereals, white rice, potatoes, rice cakes, sugary drinks.

You can lower the effect of a meal by adding **fibre, protein and fat**: toast with eggs and avocado raises blood sugar more slowly than toast with jam.

CHAPTER

Reading food labels

The Nutrition Facts table



How to read Canada's Nutrition Facts table. The US label shows the same information in a slightly different layout.

1. **Start with the serving size.** Every number below it is for that amount. If you eat twice as much, double everything.
2. **Look at calories** for that serving.
3. **Use the % Daily Value (%DV)** to judge quickly: **5% or less is a little, 15% or more is a lot.**
4. **Get less of:** saturated and trans fat, sugars, sodium.
5. **Get more of:** fibre, potassium, calcium, iron.
6. **Compare products** by making sure the serving sizes are similar.

The ingredient list

Ingredients are listed **by weight, from most to least**. If sugar or salt is in the first three, it's a big part of the food. In Canada, sugars are grouped together in brackets after the word "Sugars", so you can see all the sources at once.

Other names for sugar

Agave, brown rice syrup, cane juice, corn syrup, dextrose, fructose, glucose, high-fructose corn syrup, honey, invert sugar, maltose, maple syrup, molasses, sucrose, fruit juice concentrate.

The front-of-package symbol (Canada)

Since **January 1, 2026**, most prepackaged foods in Canada that are high in **saturated fat, sugars or sodium** must show a black-and-white magnifying-glass symbol on the front saying "High in". It's a quick warning sign: if you see it, check the table.

Allergens

Canadian labels must clearly declare the priority food allergens, gluten sources and added sulphites, either in the ingredient list or in a "Contains" statement:

Eggs · milk · mustard · peanuts · crustaceans and molluscs · fish · sesame seeds · soy · sulphites · tree nuts · wheat and triticale (and gluten from barley, oats, rye).

"May contain" statements are voluntary warnings about possible cross-contact.

Claims, decoded

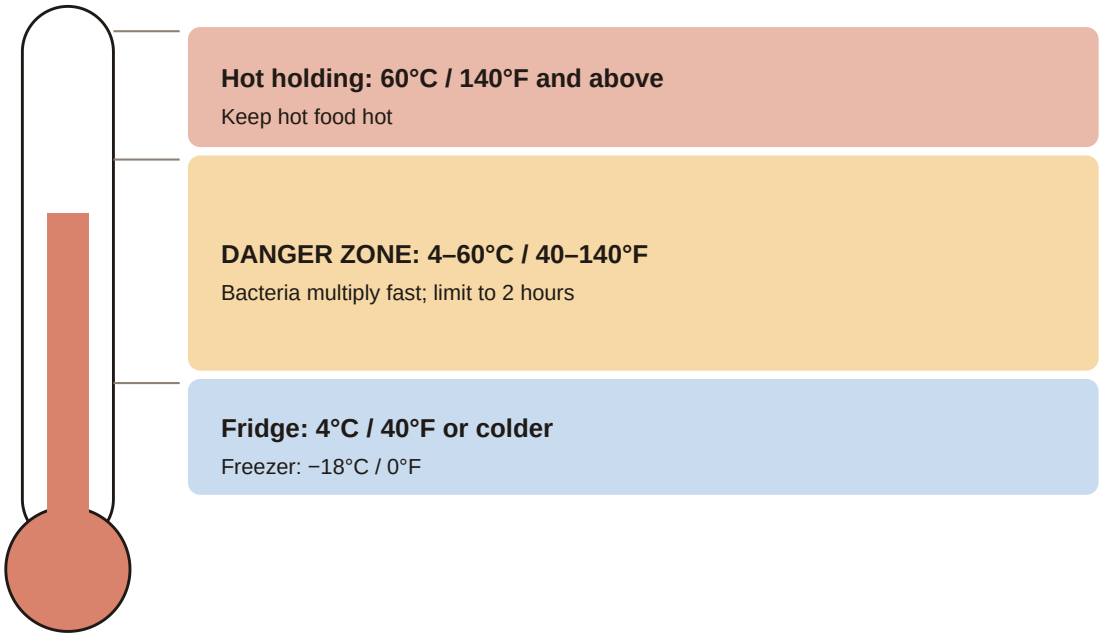
CLAIM	WHAT IT MEANS
"Source of fibre"	At least 2 g of fibre per serving
"High source of fibre"	At least 4 g per serving
"Low in sodium"	140 mg or less per serving
"Reduced in…"	At least 25% less than the regular version, but it can still be high
"Natural"	Not a nutrition claim. Says nothing about health
"Multigrain"	Several grains, but not necessarily whole grains. Look for "whole grain" first in the ingredients

CHAPTER

Food safety at home

Most foodborne illness at home comes from a few mistakes. Four habits prevent most of them: **clean, separate, cook, chill.**

The danger zone



Bacteria grow fastest between 4°C and 60°C (40°F–140°F).

- Don't leave perishable food at room temperature for more than **2 hours** (1 hour if it's very hot, above 32°C / 90°F).
- Keep your fridge at **4°C (40°F) or colder** and your freezer at **−18°C (0°F)**. Use a fridge thermometer.
- Thaw food in the fridge, in cold water or in the microwave, not on the counter.

Safe internal cooking temperatures

Use a food thermometer in the thickest part. Colour isn't a reliable test. These are Health Canada's guidelines:

FOOD	COOK TO
Poultry pieces (chicken, turkey, duck)	74°C (165°F)
Whole poultry	82°C (180°F)
Ground meat: beef, pork, lamb, veal	71°C (160°F)
Ground poultry	74°C (165°F)
Beef, veal, lamb steaks and roasts	63°C (145°F) medium-rare; 71°C (160°F) medium; 77°C (170°F) well done
Pork pieces and whole cuts	71°C (160°F)
Fish	70°C (158°F)
Egg dishes	74°C (165°F)
Leftovers, stuffing, hot dogs	74°C (165°F)

Clean and separate

- Wash hands with soap for 20 seconds before cooking and after handling raw meat, poultry, fish or eggs.
- Use separate cutting boards for raw meat and for ready-to-eat food.
- Don't rinse raw chicken; it splashes bacteria around the sink.
- Store raw meat on the **bottom shelf** of the fridge in a container, so it can't drip on other food.

How long food keeps in the fridge

FOOD	FRIDGE (4°C)
Raw ground meat and poultry	1–2 days
Raw chicken pieces	2–3 days
Raw steaks, chops and roasts	3–5 days
Raw fish	1–2 days
Cooked leftovers	2–4 days
Cooked rice	1–2 days; cool quickly and refrigerate within 1 hour
Opened deli meats	3–4 days
Hard-boiled eggs	7 days

When in doubt, throw it out. Most frozen food stays safe indefinitely at −18°C; quality is best within a few months.

CHAPTER

Myths and facts

MYTH	WHAT THE EVIDENCE SAYS
Carbs make you gain weight	Excess calories from any source do. Whole carbs with fibre are part of most healthy diets
Eating fat makes you fat	Fat is calorie-dense, but healthy fats are essential and help you feel full
Brown sugar and honey are much healthier than white sugar	They're all sugars. The tiny amounts of minerals don't make a real difference
You need to detox	Your liver and kidneys do that. Detox teas and juices aren't needed
Eating late at night causes weight gain	Total intake across the day matters more than the clock
Frozen vegetables are less nutritious	They're frozen soon after picking and are often just as nutritious as fresh
Eggs are bad for your heart	For most healthy people, eggs can be part of a balanced diet
Gluten-free is healthier	Only for people with coeliac disease or gluten sensitivity. Many gluten-free products are more refined
Skipping breakfast slows your metabolism	It doesn't. Eat breakfast if it helps you, skip it if it doesn't

A note on special diets: if you have diabetes, kidney disease, high blood pressure, food allergies or another medical condition, a registered dietitian can give advice tailored to you. In Canada, many provinces offer free dietitian advice by phone.

CHAPTER

Your kitchen spice guide

Spices add flavour without salt, sugar or calories.

SPICE	FLAVOUR	TRY IT IN
Turmeric	Warm, earthy, golden colour	Curries, lentils, scrambled eggs, rice
Cumin	Smoky, nutty	Chilli, tacos, roasted vegetables, dals
Coriander seed	Citrusy, gentle	Curries, marinades, pickles
Cinnamon	Sweet, warm	Oats, baking, stews, coffee
Paprika (smoked)	Smoky, sweet	Roast chicken, potatoes, eggs
Garam masala	Warm blend	Finish curries, roast vegetables
Black pepper	Sharp heat	Almost everything; helps turmeric absorption
Ginger	Zesty heat	Stir-fries, tea, soups, baking
Curry leaves	Aromatic, nutty	South Indian tempering for dals and coconut dishes
Chilli flakes	Heat	Pasta, pizza, stir-fries
Oregano	Earthy, herbal	Tomato sauces, Greek salads, roast vegetables
Garlic powder	Savoury	Rubs, dressings, roasted vegetables